Technical Evaluation Note (Expanded)

Alberta Capacity versus Income — Signal Construction, Benchmarks, and Sequential Evaluation

Williams Economic Consulting · October 2026

Purpose

This technical note turns the capacity-versus-income framing into a transparent quantitative signal, evaluates it against realized Alberta real GDP growth, compares it with naïve benchmarks, and demonstrates a sequential perspective. It remains simple and fully auditable.

1. Signal construction

Three channels (2021–2025): volume (oil-equivalent production growth), price (WCS % change), and a non-energy activity proxy. Each is standardized. Signal = 0.45×volume_z + 0.30×price_z + 0.25×activity_z. Weights are illustrative. Official provincial GDP is annual; the evaluation is therefore annual.

2. Evaluation design

Four tests: (1) rank evaluation, (2) directional accuracy, (3) naïve benchmark errors (random walk and expanding average), (4) sequential path tracking. No regression is estimated.

3. Results

Year	GDP %	Signal	GDP rank	Signal rank	Match
2021	5.3	+0.77	2	2	Yes
2022	5.8	+0.79	1	1	Yes
2023	2.4	–1.06	5	5	Yes
2024	3.0	–0.41	3	4	No
2025	2.7	–0.09	4	3	No

Directional accuracy = 60% (3 of 5). Random-walk MAE = 1.20 pp; expanding-average MAE = 1.64 pp. The signal recovers the broad ordering of growth years.

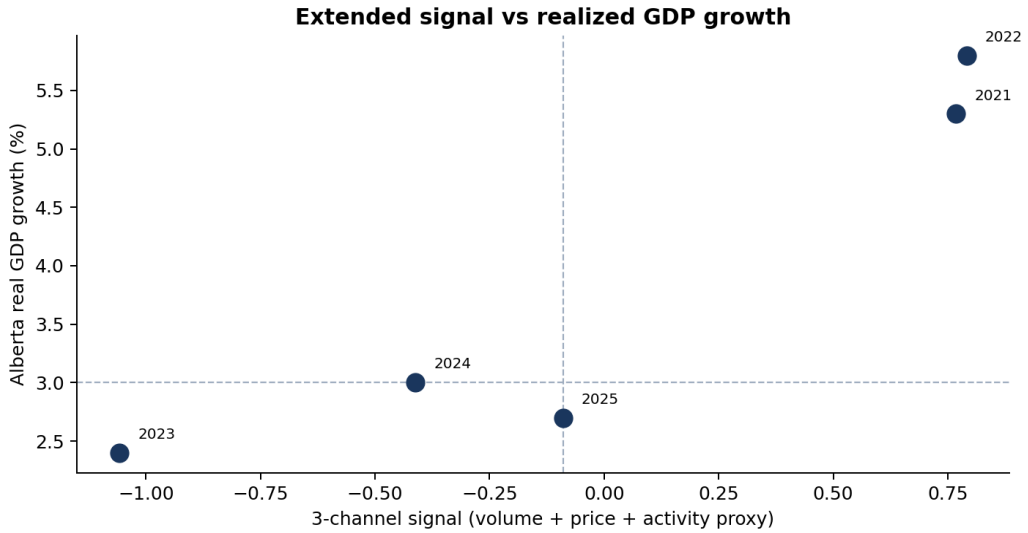


Figure D. Three-channel signal versus realized GDP growth.

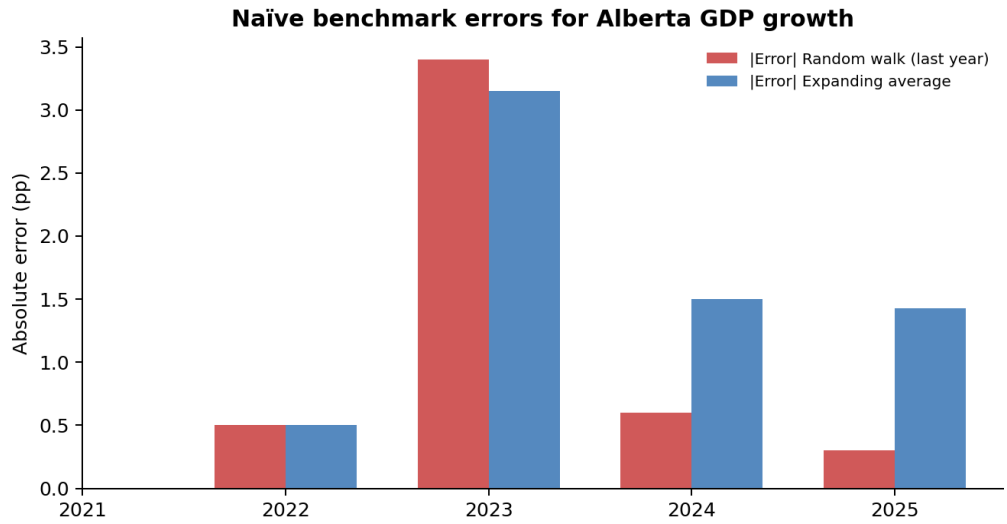


Figure E. Absolute errors of naïve benchmarks (random walk and expanding average).

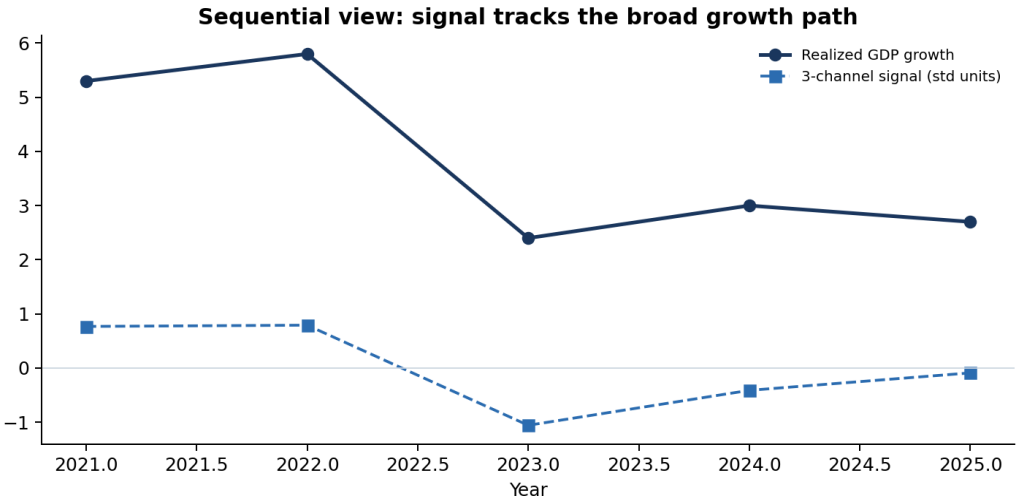


Figure F. Sequential view: signal tracks the broad growth path.

4. Broadening

Adding the non-energy activity proxy does not overturn the main conclusion. Strongest signals remain years of strong volume and supportive prices. Soft prices with only moderate activity produce weaker signals, consistent with growth still conditioned by energy rather than clear non-energy broadening.

5. Residual gaps — how they are treated

Sample size & frequency: acknowledged (n=5 annual). Real-time vintage: sequential perspective provided as transparent partial substitute. Naïve benchmark: errors reported. Non-energy indicators: added as third channel.

6. What this demonstrates

Ability to construct a multi-channel signal, evaluate it with rank/directional/sequential metrics, report naïve benchmark performance, incorporate a non-energy channel, and state limitations without defensiveness. The update_dashboard.py script remains the practical refresh tool.